



Relation-Guided and Patient-Adaptive Hypernetworks for Multi-Task Assessment in Corneal Confocal Microscopy

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ABSTRACT

Corneal confocal microscopy (CCM) provides a non-invasive window into ocular and systemic neuroimmune alterations, but clinical translation is hindered by manual interpretation and fragmented workflows. Existing deep-learning methods focus on isolated tasks, providing limited support for unified patient-level assessment. Furthermore, unified assessment faces challenges like cross-center variability, visually ambiguous phenotypes, and the need for patient-specific outcome mappings. To address these challenges, we propose a unified prior-guided and patient-adaptive framework for CCM imagery assessment. It comprises: (1) relation-guided task-specific feature extraction to retain clinically robust visual factors; (2) patient-semantic-guided feature contextualization to resolve visually ambiguous phenotypes; and (3) patient-task-conditioned hypernetworks that adapt predictive mappings to specific patient states and task identities. Together, these components unify ocular and systemic neuropathy diagnosis, continuous metric estimation, and both short- and long-term prognoses. Across 35,065 CCM images from eight centers in three countries, the framework achieved superior performance on diagnostic, metric-regression, and prognostic tasks, with consistent generalization to independent external cohorts. Beyond predictive accuracy, the estimated metrics preserved established clinical relationships with dry eye disease severity, short-term outcomes, and diabetic peripheral neuropathy status. These findings highlight the potential of CCM-derived digital biomarkers for integrated disease phenotyping. The source code is publicly available at <https://github.com/LHS-A/CCM-DiagProg>.

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1. Introduction

The cornea serves as an accessible window into neuroimmune health, with corneal confocal microscopy (CCM) en-

abling non-invasive, cellular-resolution visualization of the sub-basal nerve plexus and resident immune cells (Gad *et al.*, 2022; Alam *et al.*, 2022). This microscopic phenotyping provides the biological evidence necessary to stratify disorders that share overlapping symptoms but are driven by distinct mechanisms. As visualized in Fig. 1, healthy corneas exhibit a dense, uniform nerve network. In contrast, dry eye disease (DED) is driven by inflammatory activation, marked by the prominent infiltration of immune cells (Langerhans cells). Neuropathic corneal pain (NCP) presents a structurally decoupled phenotype defined by localized neural aberrations, such as microneuromas, stemming from somatosensory dysfunction (Dieckmann *et al.*, 2017; Moein *et al.*, 2020). Beyond localized ocular conditions, CCM also non-invasively tracks systemic neuropathies such as diabetic peripheral neuropathy (DPN). Here, the transition from Type 1 diabetes mellitus without diabetic peripheral neuropathy (DPN-) to Type 1 diabetes mellitus with diabetic peripheral neuropathy (DPN+) is characterized by profound, progressive corneal nerve loss rather than isolated immune or neuroma events. When interpreted jointly with metabolic markers like HbA1c, these distinct structural phenotypes reliably stratify neuro-metabolic risk (Braffett *et al.*, 2020).

To provide comprehensive patient management, however, these microscopic phenotypes must be translated beyond disease identification into quantitative clinical assessment and future outcome prediction. For ocular surface diseases, structural alterations are associated with functional endpoints such as tear-film break-up time (TBUT), corneal fluorescein staining (CFS), tear secretion (SIT), and patient-reported symptoms (OSDI) (Wolffsohn *et al.*, 2017). Their longitudinal evolution further reflects short-term clinical outcomes and postoperative recovery, motivating the use of baseline CCM for short-term treatment and longer-term postoperative prognosis. For systemic neuropathy, joint interpretation with metabolic markers such as HbA1c provides complementary information for neuro-metabolic risk assessment (Braffett *et al.*, 2020). Despite these biological links, current assessment remains fragmented: CCM interpretation is largely manual (Alam *et al.*, 2022), while complementary functional and metabolic measurements require separate contact-based, dye-assisted, questionnaire-based, or blood-based procedures. A unified framework that maps CCM phenotypes to diagnosis, continuous clinical metrics, and longitudinal outcomes could therefore provide a more integrated patient-level assessment.

Recent deep-learning approaches for CCM have predominantly targeted isolated structural segmentation, *e.g.*, corneal nerve and Langerhans cell, or end-to-end disease classification (Li *et al.*, 2025b,c,a; Mou *et al.*, 2022; Zhao *et al.*, 2020; Koseoglu *et al.*, 2025; Preston *et al.*, 2022; Chen *et al.*, 2024). While demonstrating technical feasibility, these methods remain largely confined to isolated anatomical structures or individual clinical objectives, limiting unified patient-level assessment. To the best of our knowledge, no prior automated CCM study has directly regressed clinically established continuous endpoints (CFS, TBUT, SIT, OSDI, or HbA1c). Furthermore, no framework integrates complex DED, NCP and DPN stratification, continuous metric regression, and longitudinal prognos-

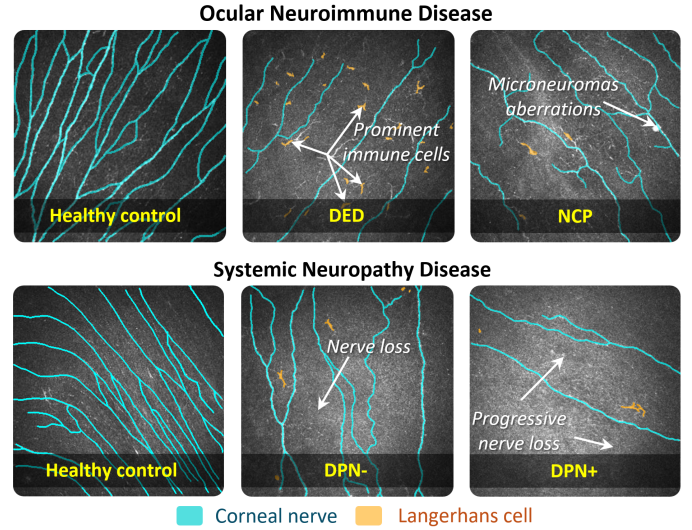


Fig. 1. Distinct microscopic imaging characteristics of ocular neuroimmune and systemic neuropathic diseases via CCM.

tic assessment (both short-term treatment and long-term postoperative) within a single framework. Consequently, current systems remain fragmented across isolated tasks and provide limited support for unified patient-level assessment spanning disease discrimination, quantitative clinical characterization, and longitudinal monitoring.

Developing this much-needed unified framework introduces two critical computational bottlenecks. First, *task conflicts and acquisition/cohort shifts destabilize learned features*. Categorical risk stratification, continuous regression, and prognosis necessitate heterogeneous visual evidence. Stratification requires invariant structural boundaries, while regression demands high sensitivity to localized morphological changes. Forcing these divergent objectives into a shared feature space induces gradient conflicts and negative transfer. Unmitigated acquisition variability and cross-center cohort shifts exacerbate this instability, obscuring mechanism-grounded neuroimmune markers behind imaging nuisance associations. Second, *rigid task heads fail to resolve patient-specific functional mappings*. Conventional multi-head or Mixture-of-Experts (MoE) architectures restrict sample-dependent routing to a finite set of fixed configurations, failing to dynamically adapt the prediction function to the patient's unique clinical state and objective. Overcoming these barriers requires a robust architecture capable of structurally decoupling visual evidence to resolve task conflicts, contextualizing this evidence with the patient's semantic records, and dynamically generating customized prediction parameters.

To address these clinical and computational challenges, we propose a unified, patient-adaptive framework for CCM-based ocular neuroimmune and systemic neuropathy assessment. Our approach spans early warning risk stratification, continuous clinical metric regression, and longitudinal prognosis by progressively resolving heterogeneity across features, patient contexts, and prediction functions. First, relation-guided task-specific feature extraction organizes latent visual dependencies using clinically induced relations, and applies HSIC/KCI-based coarse-to-fine filtering. This allows each objective to

retain its own target-relevant and conditionally robust visual factors rather than relying on a rigid shared representation. Second, patient-semantic-guided feature contextualization integrates patient-specific clinical data to reinterpret visual evidence, thereby complementing visually similar CCM phenotypes with patient-specific clinical context while preserving an image-only pathway when such information is incomplete. Third, a patient-task-conditioned hypernetwork dynamically generates custom prediction parameters for every patient-task pair. By replacing static population-level task heads with dynamically conditioned functions, the framework adapts to both heterogeneous clinical objectives and individual patient states. This enables the final mapping to adapt not only across heterogeneous objectives but also across patient states within the same objective.

The main contributions of this work are threefold:

- To our knowledge, this is the first automated CCM framework to jointly execute neuroimmune and systemic neuropathy diagnosis, alongside continuous functional metric regression, short-term treatment and long-term postoperative prognosis. By structurally decoupling multi-view features, it achieves precise stratification of healthy controls, DED, NCP, and DPN.
- The framework learns patient-adaptive mappings to predict established clinical reference endpoints (CFS, TBUT, SIT, OSDI, HbA1c) from microscopic CCM phenotypes. This reduces reliance on fragmented, contact-based measurements and enables objective neuro-metabolic risk tracking.
- We introduce a dynamic architecture that moves beyond fixed feature sharing. By combining clinically induced relation-guided representation learning, patient-semantic contextualization, and a conditioned hypernetwork, the framework effectively resolves task interference, adapts to patient heterogeneity, and mitigates cross-center representation shifts.

2. Methods

Fig. 2 illustrates the proposed unified architecture. To address task conflicts and acquisition/cohort variability across heterogeneous CCM objectives, we first construct clinically induced visual relations and apply task-specific feature filtering to decouple task-relevant visual evidence. Because visually similar CCM phenotypes may correspond to different patient states, the selected evidence is then contextualized with task-aware patient semantics. Finally, to move beyond fixed prediction mappings across tasks and patients, a shared hypernetwork jointly conditioned on patient semantics and task identity dynamically generates prediction parameters for each patient-task pair.

2.1. Relation-guided task-specific feature extraction

The first challenge is to learn distinct task-relevant evidence within a shared visual encoder while limiting nuisance relations induced by acquisition or cohort differences. We therefore address this problem in two successive stages. First, we organize the overall inter-channel relation structure using a clinically induced relation prior, such that the shared visual representation preferentially preserves relations supported by clinical evidence. Second, because a clinically regularized feature space

may still contain channels irrelevant to a particular objective, we identify a compact task-specific subset through coarse-to-fine statistical filtering. Accordingly, the first stage models and clinically aligns visual relations, whereas the second performs task-specific robust feature filtering.

2.1.1. Visual relation representation

We first characterize the relation structure encoded by the visual backbone before introducing clinical guidance. For the n -th CCM image I_n , the ResNet-50 encoder E_{vis} produces $F_n = E_{\text{vis}}(I_n) \in \mathbb{R}^{C \times H \times W}$, where C denotes the number of feature channels and $H \times W$ the spatial resolution. Let F_n^i denote the activation map of channel i . Because channel activation ranges may differ substantially, direct comparison could bias relation estimates toward high-magnitude channels. We first apply min-max normalization to each activation map and denote the normalized result by $\bar{F}_n^i$.

Each channel captures the spatial response of a learned visual pattern. Individual channel responses, however, do not reveal whether two patterns co-activate at similar spatial locations. We therefore model pairwise channel relations according to the spatial similarity of their normalized activation maps. Let $\bar{\mathbf{f}}_n^i \in \mathbb{R}^{HW}$ denote the vectorized form of $\bar{F}_n^i$. For the current image mini-batch $\mathcal{B}_{\text{img}}$, the feature relation modeling block computes

$$M_{\text{vis}}(i, j) = \frac{1}{|\mathcal{B}_{\text{img}}|} \sum_{n \in \mathcal{B}_{\text{img}}} \frac{\langle \bar{\mathbf{f}}_n^i, \bar{\mathbf{f}}_n^j \rangle}{\|\bar{\mathbf{f}}_n^i\|_2 \|\bar{\mathbf{f}}_n^j\|_2 + \epsilon} \quad (1)$$

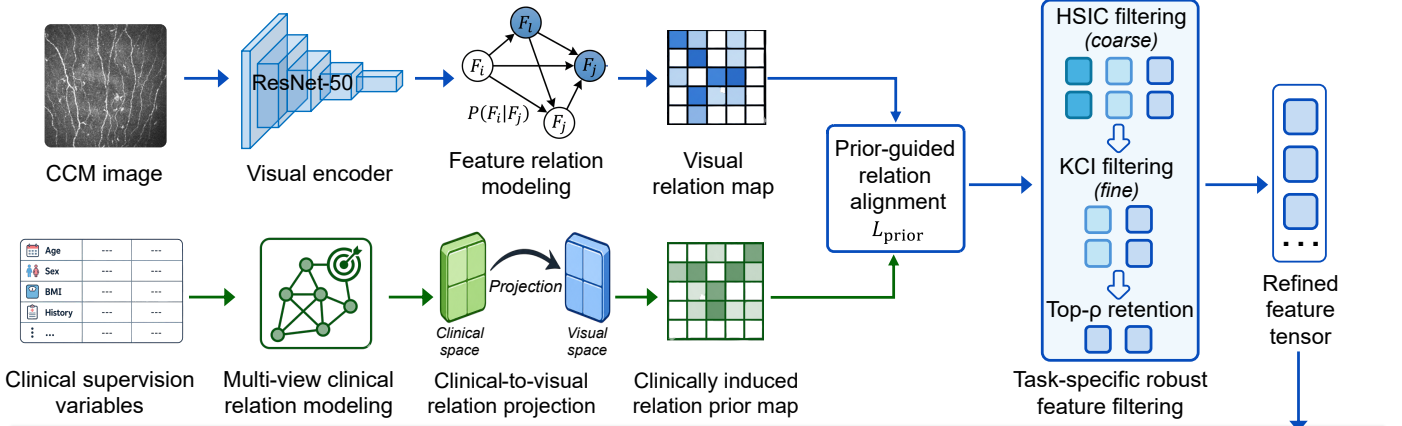
A large $M_{\text{vis}}(i, j)$ therefore indicates consistent spatial co-activation of channels i and j across the mini-batch. The resulting $M_{\text{vis}} \in \mathbb{R}^{C \times C}$ summarizes the inter-channel relation structure currently encoded by the visual backbone. M_{vis} , however, cannot distinguish clinically meaningful phenotype relations from incidental associations induced by imaging or cohort variation. We therefore construct an independent relation reference from clinical variables to guide the visual relation space.

2.1.2. Clinically induced relation alignment

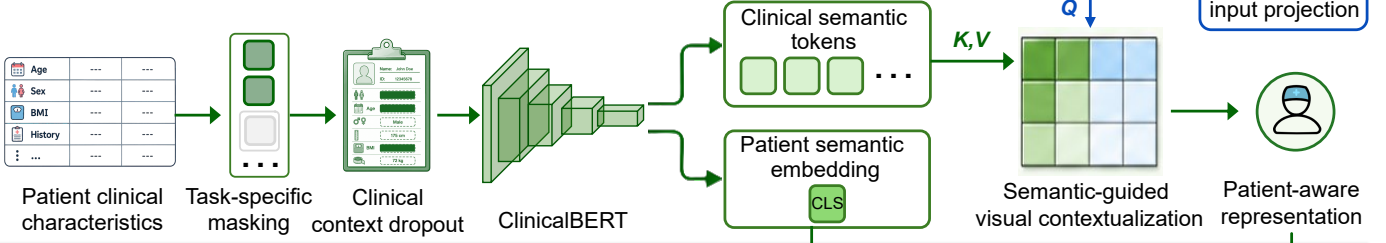
To distinguish clinically supported relations from nuisance image associations, we derive an external relation reference from the available clinical variables. These variables provide complementary information on disease state, functional measurements, and prognosis, allowing their pairwise relations to guide visual feature organization. Because clinical-variable dependencies may be linear, monotonic, nonlinear, or more general, a single relation measure is insufficient. For each prediction task, let $C_{\text{clin}} = [c^1, \dots, c^M]$ denote the M clinical supervision variables available in the corresponding training partition. Clinical variables are standardized or encoded according to their data types using statistics derived exclusively from the training partition, with each patient contributing once to clinical relation estimation.

We characterize each clinical-variable pair using four complementary relation views: absolute Pearson correlation for linear association, absolute Spearman correlation for monotonic association, distance correlation (dCor) for more general statistical dependence, and normalized mutual information (NMI)

(a) Relation-guided task-specific feature extraction



(b) Patient-semantic-guided feature contextualization



(c) Patient-task-conditioned hypernetwork prediction

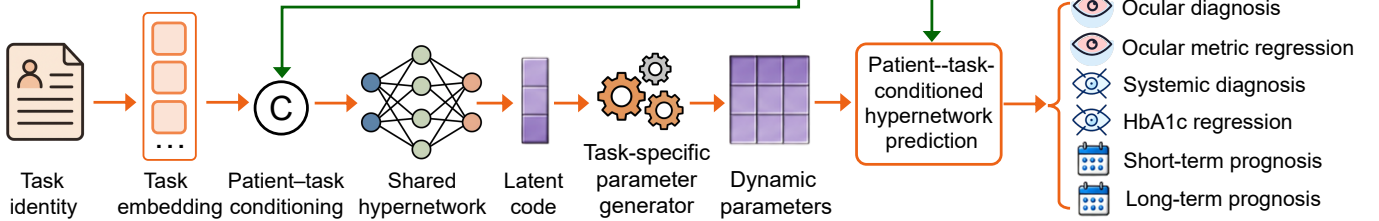


Fig. 2. Overview of the proposed framework. (a) Relation-guided task-specific feature extraction identifies task-relevant CCM evidence under clinical relation guidance. (b) Patient-semantic-guided feature contextualization integrates available patient context with the selected visual evidence. (c) Patient-task-conditioned hypernetwork prediction dynamically adapts the prediction function to each patient-task pair.

for shared information. Pearson and Spearman correlations are used for ordered numerical variables, whereas dCor and NMI accommodate heterogeneous variable types. For a clinical-variable pair (c^m, c^r) , the applicable relation strengths are collected as

$$\mathbf{a}_{m,r} = [\rho_P(c^m, c^r), \rho_S(c^m, c^r), \text{dCor}(c^m, c^r), \text{NMI}(c^m, c^r)]^T \quad (2)$$

where only statistically applicable views are retained for each variable pair. We use relation magnitude rather than correlation direction because the visual relation map in Eq. (1) represents non-negative similarity strength. Because the four views may differ in reliability, their stability is estimated by bootstrap re-sampling. Let v_k denote the mean bootstrap variance of relation view k over the clinical-variable pairs for which that view is applicable. A more stable view receives a larger weight, $\alpha_k = \exp(-v_k/\tau_v) / \sum_{\ell} \exp(-v_{\ell}/\tau_v)$, with a fixed temperature $\tau_v = 0.05$. For each variable pair, these weights are renormalized over its available views and combined to form the clinical relation matrix $A_{\text{rel}} \in \mathbb{R}^{M \times M}$. Thus, $A_{\text{rel}}(m, r)$ represents the clinically supported relation strength between c^m and c^r , with unstable estimates receiving less weight.

A_{rel} and M_{vis} cannot be compared directly because they are indexed by clinical variables and visual channels, respectively. We therefore establish a soft correspondence between the two spaces. After the visual encoder has undergone task-supervised warm-up and acquired basic task-discriminative representations, channel i of image n from patient s is summarized as $z_{n,s}^i = \text{GAP}(F_n^i)$. To avoid weighting patients according to the number of acquired CCM images, these responses are first averaged within each patient as $\bar{z}_s^i = |\mathcal{I}_s|^{-1} \sum_{n \in \mathcal{I}_s} z_{n,s}^i$, where $\mathcal{I}_s$ denotes the images of patient s . The sequence $\{\bar{z}_s^i\}$ therefore characterizes channel responses across patients, enabling association estimation with each clinical variable without repeated patient-level observations.

For each channel-clinical-variable pair (i, m) , we estimate the association strength across patients using the same applicable multi-view relation measures and bootstrap-stability weighting strategy, yielding the channel-clinical association score $R(i, m)$. The relative correspondence is normalized as $\Pi(i, m) = \exp(R(i, m)/\tau_r) / \sum_{r=1}^M \exp(R(i, r)/\tau_r)$ (fixed $\tau_r = 0.5$), and the clinical relation structure is projected into the visual channel space as $M_{\text{prior}} = \Pi A_{\text{rel}} \Pi^T \in \mathbb{R}^{C \times C}$. If two clinical vari-

ables exhibit a strong and stable relation, and two visual channels are respectively associated with those variables, the corresponding channel pair receives a stronger relation in M_{prior} . This projects clinical-variable relations into a task-specific visual relation prior. All clinical relation statistics, bootstrap weights, and clinical-to-visual correspondences are estimated exclusively from the training partition, and M_{prior} is fixed once constructed.

The visual relation map M_{vis} and the clinically induced relation prior M_{prior} are now defined over the same C visual channels. Before comparing them, we remove the diagonal self-relations and normalize the remaining structure to eliminate absolute-scale differences:

$$O(M) = M - \text{Diag}(\text{diag}(M)), \quad \widehat{M} = \frac{O(M)}{\|O(M)\|_F + \epsilon} \quad (3)$$

The prior-guided relation alignment objective is then

$$\mathcal{L}_{\text{prior}} = \left\| \widehat{M}_{\text{vis}} - \widehat{M}_{\text{prior}} \right\|_F^2 \quad (4)$$

Rather than reproducing individual clinical measurements, this objective aligns the relative inter-channel relation pattern with the clinically supported reference structure. Because M_{prior} is fixed after estimation, optimization updates the visual encoder while leaving the clinical reference unchanged. Relation alignment therefore stabilizes the shared feature structure using clinical evidence, but it does not determine which channels are most informative for each objective. We therefore further refine the aligned representation through task-specific feature filtering.

2.1.3. Task-specific robust feature filtering

To isolate the visual evidence required by each objective, robust feature filtering follows a coarse-to-fine strategy for prediction task t . HSIC first evaluates the marginal dependence between a visual channel and the task-specific target, whereas KCI evaluates whether this dependence remains after accounting for admissible non-target clinical variables. Top- ρ retention then controls the final channel budget. Because targets and clinical covariates are patient-level variables, channel descriptors are first aggregated within patients. Specifically, each channel map is summarized as $x_{n,s}^i = [\text{GAP}(F_n^i), \text{GMP}(F_n^i)]$, and the patient-level descriptor is $\bar{x}_s^i = |\mathcal{I}_s|^{-1} \sum_{n \in \mathcal{I}_s} x_{n,s}^i$. Let $X_i = \{\bar{x}_s^i\}_{s=1}^{N_p}$ denote the descriptors of channel i across the N_p training patients and Y_t the corresponding patient-level target representation for prediction task t . For multi-output regression, all target dimensions are treated jointly.

We quantify the marginal dependence between each channel and the task-specific target using the standard Hilbert–Schmidt Independence Criterion (HSIC) (Gretton et al., 2005). Statistical screening identifies the marginally target-related channels, which form the coarse candidate set S_t^{HSIC} . Marginal dependence alone may reflect associations mediated by other observed clinical variables. We therefore further evaluate conditional dependence using KCI. Let Z_t denote the admissible non-target clinical variables for prediction task t , excluding the direct prediction target, future measurements, and variables that directly define or reveal that target. For each channel in S_t^{HSIC} ,

we compute its KCI conditional dependence score with Y_t given Z_t using the standard kernel conditional independence formulation (Zhang et al., 2011). Channels that remain statistically supported after conditional screening form S_t^{KCI} . The channels supported by both stages are defined as $S_t^{\text{sig}} = S_t^{\text{HSIC}} \cap S_t^{\text{KCI}}$. Thus, HSIC provides coarse marginal screening, whereas KCI removes associations explained by admissible non-target clinical context.

To ensure a fixed feature budget across prediction tasks, we define $K_{\text{ch}} = \max(1, \lfloor \rho C \rfloor)$ with $0 < \rho \leq 1$ and retain exactly K_{ch} channels for every prediction task. If $|S_t^{\text{sig}}| \geq K_{\text{ch}}$, channels in S_t^{sig} are ranked by their KCI conditional dependence scores and the top K_{ch} form S_t . If $|S_t^{\text{sig}}| < K_{\text{ch}}$, all channels in S_t^{sig} are retained and the remaining positions are filled from the unselected channels in descending order of HSIC dependence score, ensuring the same feature budget for every task. For image sample n and prediction task t , the refined feature tensor is therefore $F_{\text{ref},n,t} = F_n[S_t, :, :] \in \mathbb{R}^{K_{\text{ch}} \times H \times W}$. Different prediction tasks may retain different channel combinations, but the feature budget is fixed to K_{ch} .

The selected channels determine *which* visual evidence is task-relevant. However, similar task-relevant CCM patterns may still have different clinical meanings across patients, motivating the use of patient context in the next stage.

2.2. Patient-semantic-guided feature contextualization

The second challenge is to resolve patient-specific ambiguity in otherwise similar CCM phenotypes. Task-specific filtering identifies relevant visual evidence but does not determine how that evidence should be interpreted for an individual patient. We therefore construct a task-safe patient representation from available clinical information and use it to contextualize the selected visual evidence, yielding a patient-aware representation.

2.2.1. Task-aware patient semantic encoding

Because the patient-semantic branch remains active during inference, it must use only information available when the requested prediction is made, unlike the clinically induced relation prior that is estimated from training data and then fixed. To enforce this requirement and prevent target leakage, we construct a task-safe patient context through *task-specific masking*. For patient s and prediction task t , we denote the available clinical characteristics by C_s^{avail} , which may include demographic information, disease history, systemic conditions, ocular characteristics, and free-text symptoms. The direct prediction target is denoted by C_t^{target} , variables unavailable at the intended prediction time by C_t^{future} , and variables that directly define or strongly reveal the requested outcome by C_t^{proxy} . The task-safe context is defined inline as $C_{s,t}^{\text{safe}} = C_s^{\text{avail}} \setminus (C_t^{\text{target}} \cup C_t^{\text{future}} \cup C_t^{\text{proxy}})$, where $\setminus$ denotes set difference, i.e., the target, future, and proxy fields are removed from the available patient information. Task-specific admissible fields are predefined before training by consensus of three senior clinicians with over 15 years of clinical experience, excluding potential proxies. Because these variables describe patient-level characteristics, all CCM images from the same patient share the same task-safe semantic context

for a given prediction task, while their visual representations remain image-specific.

After task-specific masking, clinical-context availability may still vary substantially across patients. We therefore apply *clinical context dropout* to expose the semantic branch to varying context availability during training. For a patient–task pair with $n_{s,t}^{\text{clin}}$ currently observed task-safe fields, the number of additionally removed fields is sampled as $k_{s,t} \sim \text{Uniform}\{0, \dots, n_{s,t}^{\text{clin}}\}$, after which $k_{s,t}$ fields are selected uniformly without replacement. Thus, $k_{s,t} = 0$ preserves all observed task-safe fields, whereas $k_{s,t} = n_{s,t}^{\text{clin}}$ simulates the complete absence of patient-specific clinical context.

To encode the heterogeneous clinical fields consistently, we serialize them into a standardized clinical sentence $T_{s,t}$ using a fixed linguistic template. The template consists of a natural-language scaffold with field placeholders: “A {age}-year-old {sex} patient with systemic status {systemic_status}, medical history {medical_history}, ocular status {ocular_status}, and presenting symptoms {symptom_text}.” The expressions enclosed in braces denote field placeholders and are replaced by the corresponding values during both training and inference. When a field is unavailable or removed by context dropout, its corresponding phrase is omitted from the serialized sentence. Numerical measurements are retained in their original continuous form rather than discretized. The resulting sentence is encoded by *ClinicalBERT* (Alsentzer et al., 2019) as $H_{\text{txt},s,t} = E_{\text{txt}}(T_{s,t}) \in \mathbb{R}^{L \times d_{\text{txt}}}$, where L denotes the token length and d_{txt} the text embedding dimension. The token-level representation $H_{\text{txt},s,t}$ provides the *clinical semantic tokens* for visual contextualization, whereas the corresponding [CLS] representation $h_{s,t}^{\text{cls}} \in \mathbb{R}^{d_{\text{txt}}}$ provides the *patient semantic embedding* for prediction adaptation. When no task-safe clinical field remains after masking and context dropout, the semantic encoder is bypassed and $H_{\text{txt},s,t} = \mathbf{0}$ and $h_{s,t}^{\text{cls}} = \mathbf{0}$.

2.2.2. Patient-semantic-guided visual contextualization

With task-safe patient semantics established, we use them to reinterpret the selected visual evidence for each patient. For an image sample n associated with patient s , the *semantic-guided visual contextualization* block uses visual tokens as queries and clinical semantic tokens as keys and values, allowing each visual location to retrieve patient context relevant to its interpretation. The selected feature tensor $F_{\text{ref},n,t} \in \mathbb{R}^{K_{\text{ch}} \times H \times W}$ is rearranged into spatial tokens $X_{\text{vis},n,t} = \text{Flatten}_{\text{spatial}}(F_{\text{ref},n,t})^T \in \mathbb{R}^{HW \times K_{\text{ch}}}$. Although all prediction tasks use the same channel budget K_{ch} , different tasks retain different channel identities. We therefore use a task-specific input projection $W_{\text{in},t} \in \mathbb{R}^{K_{\text{ch}} \times d_a}$ to map each task-specific channel representation into the shared attention space, yielding $X_{n,t} = X_{\text{vis},n,t} W_{\text{in},t} \in \mathbb{R}^{HW \times d_a}$.

Patient semantics are then injected through residual cross-modal attention:

$$X_{n,s,t}^{\text{ctx}} = \text{LN}[X_{n,t} + \text{Attn}(X_{n,t} W_Q, H_{\text{txt},s,t} W_K, H_{\text{txt},s,t} W_V)] \quad (5)$$

where $\text{Attn}(\cdot)$ denotes standard scaled dot-product attention, W_Q , W_K , and W_V are learnable projections, and $\text{LN}(\cdot)$ denotes layer normalization. The attention operation preserves spatial

visual tokens while incorporating patient-specific semantic context. The contextualized tokens are aggregated by average pooling as $f_{n,s,t} = \text{AvgPool}(X_{n,s,t}^{\text{ctx}}) \in \mathbb{R}^{d_a}$, yielding the *patient-aware representation* for image n of patient s . The residual connection preserves an image-based representation when semantic context is unavailable. The resulting representation is patient-aware, but its mapping to the requested outcome remains fixed unless the prediction function itself is adapted.

2.3. Patient–task-conditioned hypernetwork prediction

The third challenge is to adapt the final prediction mapping to both patient state and task identity. Although the patient-aware representation already integrates task-relevant visual evidence with patient context, a conventional task-specific head would still apply the same mapping to every patient within a task. We therefore adapt the prediction function through patient–task conditioning. Patient semantics and task identity are combined into a joint condition, which a shared hypernetwork maps to a common latent code for task-specific parameter generation.

Patient semantics encode patient state but not task identity. Each prediction task $t \in \mathcal{T}$ is therefore represented by a learnable task embedding $e_t \in \mathbb{R}^{d_e}$. The framework considers three clinical task categories (diagnosis, regression, and prognosis), comprising six prediction tasks: ocular neuroimmune diagnosis, systemic neuropathy diagnosis, ocular-surface reference metric regression, HbA1c regression, short-term treatment prognosis, and long-term postoperative prognosis. Because the prediction function depends jointly on patient state and the requested prediction task, the *patient–task conditioning* block forms $p_{s,t} = [e_t \parallel h_{s,t}^{\text{cls}}]$, where $\parallel$ denotes concatenation. The shared hypernetwork $\mathcal{H}$ maps this condition to a latent code $g_{s,t} = \mathcal{H}(p_{s,t}) \in \mathbb{R}^{d_h}$, enabling shared patient–task conditioning across tasks.

Different prediction tasks have different output spaces and dimensionalities. A lightweight *task-specific parameter generator* G_t therefore converts the shared latent code into the weight and bias of the dynamic prediction head:

$$[\text{vec}(W_{s,t}); b_{s,t}] = G_t(g_{s,t}) \quad (6)$$

where $W_{s,t} \in \mathbb{R}^{d_a \times o_t}$, $b_{s,t} \in \mathbb{R}^{o_t}$, and o_t denotes the output dimensionality of prediction task t . Thus, patients within the same task can receive different prediction parameters according to their clinical context. The generated parameters are applied to each patient-aware image representation as $\hat{y}_{n,s,t} = f_{n,s,t}^T W_{s,t} + b_{s,t}$. When patient-specific clinical information is unavailable, $h_{s,t}^{\text{cls}} = \mathbf{0}$ and the hypernetwork is conditioned by task identity alone through $p_{s,t} = [e_t \parallel \mathbf{0}]$.

2.4. Model optimization and inference

To ensure stable optimization across heterogeneous tasks, the framework is trained in two stages. Because regression targets differ in scale, each target dimension is standardized using statistics from the corresponding training partition and inverse-transformed to its original clinical unit for evaluation. To prevent patients with more CCM images from dominating training, patient-balanced sampling assigns equal sampling weight to each patient.

The visual encoder is first warmed up with direct task supervision to establish task-discriminative representations. Let $\mathcal{L}_{\text{aux}}$ denote the supervised loss, using cross-entropy for multiclass diagnosis, binary cross-entropy for binary prognosis, and mean squared error for regression. Once this representation has converged, the clinically induced relation priors are estimated from the training partition and fixed. The encoder is then refined under joint task and relation supervision as

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{aux}} + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}}, \quad (7)$$

after which the task-specific channel subsets $\{S_t\}_{t \in \mathcal{T}}$ are fixed for Stage 2.

Stage 2 focuses on patient-adaptive prediction with the visual encoder and channel subsets fixed. The semantic contextualization and hypernetwork components are jointly optimized, with clinical context dropout applied only during training. For a mini-batch, let $\mathcal{T}_{\mathcal{B}}$ denote the represented tasks, $\mathcal{P}_t^{\mathcal{B}}$ their patients, and $\mathcal{I}_{s,t}^{\mathcal{B}}$ the corresponding CCM images. To balance supervision across tasks and patients, the prediction loss is

$$\mathcal{L}_{\text{task}} = \frac{1}{|\mathcal{T}_{\mathcal{B}}|} \sum_{t \in \mathcal{T}_{\mathcal{B}}} \frac{1}{|\mathcal{P}_t^{\mathcal{B}}|} \sum_{s \in \mathcal{P}_t^{\mathcal{B}}} \frac{1}{|\mathcal{I}_{s,t}^{\mathcal{B}}|} \sum_{n \in \mathcal{I}_{s,t}^{\mathcal{B}}} \ell_t(\hat{y}_{n,s,t}, y_{s,t}), \quad (8)$$

where ℓ_t is cross-entropy for multiclass diagnosis, binary cross-entropy for binary prognosis, and mean squared error for regression. Because the hypernetwork dynamically generates patient-task-specific prediction parameters, their magnitude is additionally regularized during optimization. Let $\mathcal{Q}_{\mathcal{B}} = \{(s, t) : s \in \mathcal{P}_t^{\mathcal{B}}, t \in \mathcal{T}_{\mathcal{B}}\}$ denote the patient-task pairs in the mini-batch. The complete Stage 2 objective is

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{task}} + \frac{\lambda_{\text{hyper}}}{|\mathcal{Q}_{\mathcal{B}}|} \sum_{(s,t) \in \mathcal{Q}_{\mathcal{B}}} (\|W_{s,t}\|_F^2 + \|b_{s,t}\|_2^2). \quad (9)$$

At inference, the requested task determines the fixed channel subset used to filter each encoded CCM image. Available task-safe patient information then contextualizes the visual representation and conditions the hypernetwork for prediction. Patient-level outputs are obtained by averaging image-level probabilities for classification and prognosis and continuous predictions for regression.

3. Experiments

3.1. Datasets and experimental setup

The study comprised two cross-sectional disease domains and two longitudinal ocular settings, spanning three clinical task categories: diagnosis, clinical metric regression, and prognosis, with the latter evaluated in short-term treatment and long-term postoperative settings. Detailed dataset characteristics are summarized in Fig. 3. Internal cohorts were used for model development with patient-level cross-validation, whereas external cohorts comprised independent patients completely excluded from model development and were drawn from partially overlapping or fully distinct centers relative to the internal cohorts. The ocular-surface reference metric and HbA1c regression tasks were derived from the same regression dataset.

A Ocular neuroimmune diseases					
Task	Cohort	Centers (Non-overlapping)	Images	Patients	Details
Diagnosis	Internal	C1 (China-Beijing) C2 (China-Ningbo) C3 (China-Wenzhou)	5,425	518	• HC - 1,936 • DED - 2,094 • NCP - 1,395
	External	C4 (China-Beijing) C5 (China-Ningbo) C7 (Italy-Padova)	3,522	378	• HC - 1,158 • DED - 1,288 • NCP - 1,076
4 metrics Regression	Internal	C1 (China-Beijing)	5,758	97	• TBUT CFS SIT OSDI - per 5,758
	External	C4 (China-Beijing)	3,056	52	• TBUT CFS SIT OSDI - per 3,056
Short-term prognosis	Internal	C1 (China-Beijing)	3,518	28	• TBUT _{1M} CFS _{1M} SIT _{1M} OSDI _{1M} - per 3,518
	External	C4 (China-Beijing)	1,027	22	• TBUT _{1M} CFS _{1M} SIT _{1M} OSDI _{1M} - per 1,027
Long-term prognosis	Internal	C1 (China-Beijing)	1,629	70	• NPD - 439 • PDE - 1,190
	External	C2 (China-Ningbo)	1,378	59	• NPD - 518 • PDE - 860
B Systemic neuropathy diseases					
Task	Cohort	Centers (Non-overlapping)	Images	Patients	Details
Diagnosis	Internal	C1 (China-Beijing) C6 (China-Ningbo)	5,920	277	• HC 1,962 • DPN- 1,685 • DPN+ 2,273
	External	C8 (UK-Manchester) C7 (Italy-Padova) C3 (China-Wenzhou)	3,832	224	• HC 929 • DPN- 1,289 • DPN+ 1,614
HbA1c regression	Internal	C1 (China-Beijing)	5,758	97	• HbA1c - 5,758
	External	C4 (China-Beijing)	3,056	52	• HbA1c - 3,056

Fig. 3. Overview of datasets and clinical evaluation tasks.

The diagnosis task category encompassed both ocular and systemic assessments. The ocular diagnosis task discriminated between healthy controls (HC), dry eye disease (DED), and neuropathic corneal pain (NCP). A separate ocular-surface reference metric regression task estimated four complementary clinical endpoints—TBUT, CFS, SIT, and OSDI—from CCM images, providing quantitative functional phenotypes in addition to categorical disease discrimination. The systemic diagnosis task discriminated among HC, T1DM without DPN (DPN-), and T1DM with DPN (DPN+). HbA1c was additionally estimated as a continuous metabolic reference, allowing corneal neural phenotypes and glycemic burden to be evaluated within the same systemic assessment framework.

Baseline (T_0) CCM and clinical measurements were acquired before treatment, followed by reassessment at one month (1M). Given the baseline CCM image and clinical information available at T_0 , the model directly predicted the absolute one-month values y_{1M} of TBUT, CFS, SIT, and OSDI. Measurements acquired at 1M were used exclusively as prediction targets and were excluded from model inputs. Preoperative baseline (T_0) CCM was used to predict the six-month outcome after FS-LASIK, distinguishing non-persistent dry eye (NPD) from persistent dry eye (PDE). No postoperative information was available to the model at baseline prediction.

All CCM images were acquired using the Heidelberg Retina Tomograph III with the Rostock Cornea Module (HRT-III/RCM) and analyzed at a spatial resolution of 384×384 pixels. All data partitioning was performed globally at the patient level using unique patient identifiers: patients contributing to multiple tasks were assigned consistently to the same partition across tasks, while the fixed external cohorts contained non-overlapping patients and were not involved in model development or hyperparameter selection. The study was approved by the appropriate institutional ethics committees of the participating centers and by the Institute of Biomedical Engineering, Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences (20240320-0022R). Each center conducted the study under its corresponding local ethical approval or applicable waiver. All images and associated metadata were anonymized before analysis, and all procedures followed the Declaration of Helsinki.

3.2. Implementation details

A ResNet-50 pretrained on ImageNet was used as the visual encoder, and pretrained ClinicalBERT (Tiny) was used for patient-semantic encoding. For multi-view clinical relation modeling, normalized mutual information (NMI) was computed by normalizing mutual information by the geometric mean of the corresponding marginal entropies; nearest-neighbor estimators were used for continuous variables, whereas empirical distributions were used for purely categorical relations. For HSIC and KCI, RBF kernels were used for continuous descriptors and regression targets and a label kernel for categorical targets, with bandwidths determined by the training-partition median heuristic. Statistical significance was assessed using permutation-based testing with Benjamini–Hochberg false-discovery-rate correction, and KCI additionally used generalized cross-validation for kernel residualization. Patient-balanced sampling uniformly selected patients and then images within each patient for each task. AdamW was applied with an initial learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , cosine annealing to 1×10^{-6} , and a total batch size of 16 images. Training was limited to 500 epochs with early stopping after 20 epochs without validation improvement. The tuned hyperparameters were selected on patient-level validation splits within the training folds and fixed at $\lambda_{\text{prior}} = 1.0$, $\rho = 30\%$, and $\lambda_{\text{hyper}} = 5 \times 10^{-4}$ before held-out evaluation.

All experiments used strict patient-level partitioning and evaluation. Internal point estimates were computed from pooled out-of-fold patient predictions, whereas external point estimates were computed from patient-wise predictions averaged across the five fold-specific models. SD denotes variability across the five internal held-out folds and the five fold-specific models on the fixed external cohort, respectively. Statistical comparisons were performed on these pooled internal and ensemble external patient-level predictions using two-sided paired permutation tests for class-wise one-vs-rest AUCs and endpoint-specific MAEs, and two-sided DeLong tests for binary AUC. Bonferroni correction was applied across the corresponding comparisons. For multi-output tasks, the largest adjusted p -value determined the reported significance threshold,

such that an entry of $p < 0.005$ indicates that all corresponding outcomes remained significant below the reported threshold; for single-output tasks, the reported threshold directly reflects the corresponding adjusted p -value. All data-dependent statistics and feature-selection quantities were estimated exclusively from the corresponding training partitions. External cohorts remained completely independent during model development and were accessed only after the model configuration and hyperparameters had been fixed.

3.3. Evaluation metrics

Classification performance was evaluated using accuracy (ACC) and area under the receiver operating characteristic curve (AUC). For multi-class diagnosis, per-class ACC and class-wise AUC were computed, with the latter using a one-vs-rest scheme; macro-averages were reported across classes. For binary prognosis, a single overall ACC and AUC were reported. Regression performance was assessed using root mean square error (RMSE) and mean absolute error (MAE). All primary metrics were computed from patient-level aggregated predictions, with higher ACC/AUC and lower RMSE/MAE indicating better performance.

3.4. Comparison methods

We compared the proposed framework with eight representative medical multi-task baselines: JMMLRC (Brand *et al.*, 2020), QCNN (Lao *et al.*, 2025), C-former (Mineo *et al.*, 2024), Multi-Cog (Ma *et al.*, 2026), MM-MIL (Huang *et al.*, 2026), MM-DLS (Liu *et al.*, 2026), DenseMoE (Ding *et al.*, 2025), and KGT (Zhao *et al.*, 2026). All baselines were evaluated on the same prediction tasks using identical data partitions, patient information, and evaluation protocols.

4. Experimental Results

4.1. Quantitative comparison

Tables 1–3 compare the proposed framework with representative multi-task methods across six prediction tasks spanning three clinical task categories on both internal and independent external cohorts. Overall, the *Ours (full)* configuration achieved the best performance across diagnosis, clinical metric regression, and longitudinal prognosis, with the advantage consistently preserved under external validation.

For ocular neuroimmune diagnosis, the *Ours (full)* configuration improved the average AUC over the strongest competing method by 0.042 internally (0.950 vs. 0.908) and 0.047 externally (0.906 vs. 0.859). For systemic small-fiber neuropathy diagnosis, the corresponding average AUC gains were 0.050 internally (0.938 vs. 0.888) and 0.044 externally (0.912 vs. 0.868). For clinical metric regression, MAE decreased for each of TBUT, CFS, SIT, and OSDI on both internal and external cohorts, while HbA1c MAE decreased from 2.571 to 0.820 internally and from 3.092 to 0.965 externally. For longitudinal prediction, long-term prognosis AUC increased by 0.044 internally (0.973 vs. 0.929) and by 0.039 externally (0.916 vs. 0.877). The consistent direction of improvement across categorical, continuous, and longitudinal outcomes indicates that the benefits are

Table 1. Quantitative comparison of ocular neuroimmune assessment on the internal and external cohorts. Reported p -values are calculated against *Ours (full)* configuration. *Ours (full)* configuration uses complete task-safe-patient context; *Ours (image)* configuration denotes the image-only configuration with the semantic contextualization branch removed.

Methods	Metrics	Diagnosis					Regression					
		HC	DED	NCP	Avg.	p <	Metrics	TBUT	CFS	SIT	OSDI	p <
Internal validation												
JMMLRC	ACC	0.808 ^{+0.048}	0.824 ^{+0.048}	0.846 ^{+0.032}	0.826 ^{+0.039}		RMSE	7.247 ^{+0.234}	7.104 ^{+0.306}	7.481 ^{+0.295}	7.168 ^{+0.364}	
	AUC	0.848 ^{+0.031}	0.862 ^{+0.027}	0.887 ^{+0.030}	0.866 ^{+0.021}	.001	MAE	5.233 ^{+0.266}	5.116 ^{+0.137}	5.469 ^{+0.283}	5.082 ^{+0.156}	.001
QCNN	ACC	0.839 ^{+0.022}	0.838 ^{+0.027}	0.852 ^{+0.040}	0.843 ^{+0.024}		RMSE	6.716 ^{+0.297}	6.381 ^{+0.213}	7.126 ^{+0.266}	6.549 ^{+0.346}	
	AUC	0.879 ^{+0.023}	0.903 ^{+0.021}	0.883 ^{+0.018}	0.888 ^{+0.014}	.001	MAE	4.524 ^{+0.136}	4.097 ^{+0.224}	4.834 ^{+0.267}	4.212 ^{+0.144}	.001
C-former	ACC	0.826 ^{+0.031}	0.831 ^{+0.046}	0.885 ^{+0.030}	0.847 ^{+0.028}		RMSE	5.846 ^{+0.287}	5.723 ^{+0.155}	6.357 ^{+0.336}	6.874 ^{+0.221}	
	AUC	0.872 ^{+0.027}	0.886 ^{+0.022}	0.920 ^{+0.019}	0.893 ^{+0.021}	.001	MAE	3.414 ^{+0.114}	3.187 ^{+0.203}	3.923 ^{+0.136}	4.136 ^{+0.227}	.001
Multi-Cog	ACC	0.852 ^{+0.028}	0.854 ^{+0.033}	0.868 ^{+0.027}	0.858 ^{+0.020}		RMSE	6.129 ^{+0.184}	5.751 ^{+0.276}	6.596 ^{+0.254}	6.012 ^{+0.333}	
	AUC	0.894 ^{+0.011}	0.915 ^{+0.018}	0.897 ^{+0.022}	0.902 ^{+0.014}	.001	MAE	3.867 ^{+0.226}	3.443 ^{+0.117}	4.114 ^{+0.246}	3.576 ^{+0.131}	.001
MM-MIL	ACC	0.817 ^{+0.043}	0.815 ^{+0.052}	0.859 ^{+0.022}	0.830 ^{+0.031}		RMSE	6.984 ^{+0.315}	6.642 ^{+0.193}	7.318 ^{+0.386}	6.906 ^{+0.255}	
	AUC	0.859 ^{+0.018}	0.875 ^{+0.023}	0.904 ^{+0.023}	0.879 ^{+0.016}	.001	MAE	4.896 ^{+0.153}	4.423 ^{+0.233}	5.084 ^{+0.176}	4.653 ^{+0.245}	.001
MM-DLS	ACC	0.834 ^{+0.037}	0.842 ^{+0.041}	0.873 ^{+0.019}	0.850 ^{+0.022}		RMSE	5.857 ^{+0.176}	5.594 ^{+0.267}	6.371 ^{+0.244}	5.786 ^{+0.316}	
	AUC	0.887 ^{+0.022}	0.896 ^{+0.016}	0.891 ^{+0.027}	0.891 ^{+0.021}	.001	MAE	3.573 ^{+0.217}	3.236 ^{+0.086}	3.883 ^{+0.243}	3.347 ^{+0.124}	.001
DenseMoE	ACC	0.844 ^{+0.019}	0.856 ^{+0.023}	0.873 ^{+0.031}	0.858 ^{+0.019}		RMSE	5.636 ^{+0.264}	5.243 ^{+0.144}	7.164 ^{+0.327}	7.513 ^{+0.216}	
	AUC	0.895 ^{+0.013}	0.905 ^{+0.019}	0.902 ^{+0.012}	0.901 ^{+0.011}	.001	MAE	2.987 ^{+0.206}	3.064 ^{+0.194}	3.756 ^{+0.135}	3.127 ^{+0.223}	.001
KGT	ACC	0.847 ^{+0.030}	0.859 ^{+0.031}	0.887 ^{+0.017}	0.864 ^{+0.018}		RMSE	5.828 ^{+0.283}	5.013 ^{+0.158}	7.625 ^{+0.361}	7.247 ^{+0.244}	
	AUC	0.897 ^{+0.019}	0.910 ^{+0.016}	0.916 ^{+0.018}	0.908 ^{+0.014}	.005	MAE	2.746 ^{+0.118}	3.154 ^{+0.202}	3.621 ^{+0.141}	2.889 ^{+0.232}	.001
Ours (image)	ACC	0.872 ^{+0.043}	0.902 ^{+0.028}	0.917 ^{+0.024}	0.897 ^{+0.025}		RMSE	2.438 ^{+0.226}	2.285 ^{+0.108}	4.173 ^{+0.257}	4.526 ^{+0.148}	
	AUC	0.917 ^{+0.018}	0.931 ^{+0.021}	0.942 ^{+0.026}	0.930 ^{+0.015}	.005	MAE	1.521 ^{+0.184}	1.219 ^{+0.047}	1.933 ^{+0.098}	1.768 ^{+0.186}	.001
Ours (full)	ACC	0.901 ^{+0.032}	0.914 ^{+0.031}	0.940 ^{+0.019}	0.918 ^{+0.020}		RMSE	1.699 ^{+0.197}	1.574 ^{+0.083}	3.564 ^{+0.244}	3.773 ^{+0.122}	
	AUC	0.938 ^{+0.018}	0.950 ^{+0.011}	0.963 ^{+0.027}	0.950 ^{+0.015}	–	MAE	1.092 ^{+0.064}	0.815 ^{+0.046}	1.500 ^{+0.077}	1.348 ^{+0.163}	–
External validation												
Multi-Cog	ACC	0.781 ^{+0.049}	0.826 ^{+0.049}	0.813 ^{+0.035}	0.807 ^{+0.032}		RMSE	7.491 ^{+0.265}	6.862 ^{+0.342}	7.683 ^{+0.385}	7.128 ^{+0.426}	
	AUC	0.803 ^{+0.031}	0.805 ^{+0.028}	0.854 ^{+0.027}	0.821 ^{+0.023}	.001	MAE	5.274 ^{+0.292}	5.148 ^{+0.161}	5.345 ^{+0.314}	5.267 ^{+0.169}	.001
JMMLRC	ACC	0.802 ^{+0.030}	0.847 ^{+0.024}	0.839 ^{+0.036}	0.829 ^{+0.024}		RMSE	6.762 ^{+0.333}	6.427 ^{+0.196}	6.985 ^{+0.441}	6.758 ^{+0.289}	
	AUC	0.837 ^{+0.031}	0.835 ^{+0.028}	0.847 ^{+0.020}	0.840 ^{+0.019}	.001	MAE	4.789 ^{+0.160}	4.214 ^{+0.231}	4.962 ^{+0.185}	4.463 ^{+0.253}	.001
MM-DLS	ACC	0.793 ^{+0.036}	0.820 ^{+0.041}	0.875 ^{+0.016}	0.829 ^{+0.025}		RMSE	6.038 ^{+0.190}	5.619 ^{+0.283}	6.542 ^{+0.308}	6.953 ^{+0.376}	
	AUC	0.825 ^{+0.036}	0.841 ^{+0.015}	0.868 ^{+0.023}	0.845 ^{+0.017}	.001	MAE	3.571 ^{+0.231}	3.385 ^{+0.108}	4.027 ^{+0.252}	4.456 ^{+0.137}	.001
C-former	ACC	0.819 ^{+0.035}	0.838 ^{+0.026}	0.862 ^{+0.029}	0.840 ^{+0.022}		RMSE	6.145 ^{+0.319}	5.746 ^{+0.178}	6.418 ^{+0.423}	6.172 ^{+0.268}	
	AUC	0.848 ^{+0.025}	0.846 ^{+0.027}	0.864 ^{+0.014}	0.853 ^{+0.017}	.001	MAE	4.121 ^{+0.138}	3.529 ^{+0.229}	4.293 ^{+0.159}	3.762 ^{+0.248}	.001
QCNN	ACC	0.784 ^{+0.037}	0.811 ^{+0.032}	0.828 ^{+0.041}	0.808 ^{+0.029}		RMSE	7.174 ^{+0.241}	6.558 ^{+0.323}	7.419 ^{+0.355}	6.912 ^{+0.422}	
	AUC	0.814 ^{+0.033}	0.834 ^{+0.029}	0.844 ^{+0.021}	0.831 ^{+0.021}	.001	MAE	4.962 ^{+0.278}	4.685 ^{+0.142}	5.118 ^{+0.301}	4.921 ^{+0.164}	.001
KGT	ACC	0.796 ^{+0.043}	0.843 ^{+0.031}	0.854 ^{+0.024}	0.831 ^{+0.025}		RMSE	6.056 ^{+0.298}	5.443 ^{+0.189}	6.364 ^{+0.398}	5.941 ^{+0.262}	
	AUC	0.855 ^{+0.012}	0.838 ^{+0.016}	0.857 ^{+0.026}	0.850 ^{+0.014}	.001	MAE	3.727 ^{+0.141}	3.446 ^{+0.211}	3.948 ^{+0.164}	3.675 ^{+0.232}	.001
DenseMoE	ACC	0.819 ^{+0.024}	0.847 ^{+0.020}	0.849 ^{+0.030}	0.838 ^{+0.020}		RMSE	5.858 ^{+0.182}	4.894 ^{+0.271}	7.305 ^{+0.261}	7.642 ^{+0.364}	
	AUC	0.833 ^{+0.028}	0.842 ^{+0.022}	0.865 ^{+0.010}	0.847 ^{+0.011}	.001	MAE	3.126 ^{+0.222}	3.392 ^{+0.100}	3.381 ^{+0.243}	3.052 ^{+0.124}	.001
MM-MIL	ACC	0.816 ^{+0.027}	0.852 ^{+0.034}	0.860 ^{+0.020}	0.843 ^{+0.021}		RMSE	5.971 ^{+0.204}	4.682 ^{+0.288}	7.593 ^{+0.282}	7.424 ^{+0.401}	
	AUC	0.853 ^{+0.027}	0.855 ^{+0.014}	0.869 ^{+0.021}	0.859 ^{+0.013}	.005	MAE	2.869 ^{+0.232}	3.041 ^{+0.115}	2.858 ^{+0.251}	3.393 ^{+0.136}	.001
Ours (image)	ACC	0.856 ^{+0.023}	0.888 ^{+0.016}	0.893 ^{+0.039}	0.879 ^{+0.021}		RMSE	2.486 ^{+0.219}	1.872 ^{+0.082}	4.219 ^{+0.234}	4.683 ^{+0.161}	
	AUC	0.872 ^{+0.025}	0.866 ^{+0.048}	0.894 ^{+0.011}	0.877 ^{+0.020}	.005	MAE	1.815 ^{+0.088}	1.364 ^{+0.012}	1.937 ^{+0.086}	1.834 ^{+0.171}	.001
Ours (full)	ACC	0.871 ^{+0.028}	0.906 ^{+0.024}	0.922 ^{+0.031}	0.900 ^{+0.022}		RMSE	1.810 ^{+0.102}	1.145 ^{+0.017}	3.428 ^{+0.128}	3.958 ^{+0.243}	
	AUC	0.898 ^{+0.032}	0.899 ^{+0.029}	0.920 ^{+0.016}	0.906 ^{+0.016}	–	MAE	1.232 ^{+0.173}	0.829 ^{+0.041}	1.206 ^{+0.168}	1.117 ^{+0.054}	.001

not confined to a single endpoint. The retained advantages on independent external cohorts further support cross-center generalization.

4.2. Qualitative analysis

Fig. 4 examines model attention to clinically meaningful CCM patterns together with complementary neuro-morphometric changes. Overall, our method shows more concentrated activation over disease-relevant nerve and immune structures and less attention to visually salient but clinically less informative background regions.

In Fig. 4(a), compared with MM-DLS, Multi-Cog, DenseMoE, and KGT, our Grad-CAM responses are more spatially aligned with the clinician-annotated regions across HC, DED, and NCP examples. The difference is most evident in abnormal cases, where competing methods often produce broader or fragmented activations, whereas our model concentrates more strongly on localized nerve alterations. This suggests that the learned representation relies more on disease-related structural evidence rather than non-specific image appearance.

Fig. 4(b) further characterizes clinically relevant corneal nerve loss in the systemic neuropathy cohorts. Nerves from the held-out internal test sets and complete external cohort were segmented using CCM-Pro (Li et al., 2025b), from which

Table 2. Quantitative comparison of systemic neuropathy assessment on the internal and external cohorts. Reported p -values are calculated against *Ours (full)* configuration.

Methods	Metrics	Diagnosis				Regression			
		HC	DPN−	DPN+	Avg.	p <	Metrics	HbA1c	p <
Internal validation									
JMMLRC	ACC	0.814 _{±0.041}	0.783 _{±0.037}	0.846 _{±0.034}	0.814 _{±0.028}		RMSE	7.286 _{±0.246}	
	AUC	0.858 _{±0.013}	0.834 _{±0.028}	0.872 _{±0.011}	0.855 _{±0.014}	.001	MAE	5.274 _{±0.259}	.001
QCNN	ACC	0.840 _{±0.033}	0.797 _{±0.032}	0.858 _{±0.030}	0.832 _{±0.022}		RMSE	6.733 _{±0.317}	
	AUC	0.880 _{±0.019}	0.851 _{±0.024}	0.887 _{±0.008}	0.873 _{±0.013}	.001	MAE	4.614 _{±0.136}	.001
C-former	ACC	0.831 _{±0.028}	0.825 _{±0.034}	0.863 _{±0.021}	0.840 _{±0.022}		RMSE	5.974 _{±0.192}	
	AUC	0.872 _{±0.022}	0.884 _{±0.006}	0.880 _{±0.022}	0.879 _{±0.013}	.001	MAE	3.453 _{±0.213}	.001
Multi-Cog	ACC	0.853 _{±0.018}	0.806 _{±0.038}	0.874 _{±0.024}	0.844 _{±0.021}		RMSE	6.117 _{±0.306}	
	AUC	0.892 _{±0.016}	0.864 _{±0.010}	0.899 _{±0.007}	0.885 _{±0.009}	.001	MAE	3.896 _{±0.127}	.001
MM-MIL	ACC	0.821 _{±0.039}	0.790 _{±0.034}	0.852 _{±0.033}	0.821 _{±0.026}		RMSE	6.976 _{±0.231}	
	AUC	0.867 _{±0.011}	0.845 _{±0.024}	0.881 _{±0.010}	0.864 _{±0.012}	.001	MAE	4.947 _{±0.247}	.001
MM-DLS	ACC	0.837 _{±0.025}	0.814 _{±0.036}	0.867 _{±0.018}	0.839 _{±0.021}		RMSE	5.914 _{±0.299}	
	AUC	0.884 _{±0.018}	0.860 _{±0.011}	0.893 _{±0.019}	0.879 _{±0.013}	.001	MAE	3.586 _{±0.122}	.001
DenseMoE	ACC	0.851 _{±0.026}	0.809 _{±0.024}	0.872 _{±0.023}	0.844 _{±0.016}		RMSE	4.996 _{±0.193}	
	AUC	0.888 _{±0.005}	0.867 _{±0.018}	0.895 _{±0.006}	0.883 _{±0.008}	.001	MAE	2.853 _{±0.214}	.001
KGT	ACC	0.857 _{±0.018}	0.821 _{±0.032}	0.883 _{±0.024}	0.854 _{±0.019}		RMSE	4.832 _{±0.285}	
	AUC	0.886 _{±0.016}	0.884 _{±0.020}	0.893 _{±0.018}	0.888 _{±0.015}	.005	MAE	2.571 _{±0.109}	.001
Ours (image)	ACC	0.902 _{±0.028}	0.850 _{±0.023}	0.921 _{±0.032}	0.891 _{±0.022}		RMSE	2.148 _{±0.196}	
	AUC	0.931 _{±0.016}	0.904 _{±0.024}	0.934 _{±0.019}	0.923 _{±0.013}	.005	MAE	1.439 _{±0.154}	.001
Ours (full)	ACC	0.912 _{±0.020}	0.882 _{±0.034}	0.934 _{±0.016}	0.909 _{±0.018}		RMSE	1.276 _{±0.072}	
	AUC	0.938 _{±0.021}	0.929 _{±0.008}	0.946 _{±0.022}	0.938 _{±0.013}	−	MAE	0.820 _{±0.137}	−
External validation									
C-former	ACC	0.795 _{±0.045}	0.790 _{±0.037}	0.826 _{±0.038}	0.804 _{±0.030}		RMSE	7.428 _{±0.259}	
	AUC	0.844 _{±0.030}	0.814 _{±0.028}	0.818 _{±0.027}	0.825 _{±0.022}	.001	MAE	5.286 _{±0.279}	.001
MM-MIL	ACC	0.835 _{±0.027}	0.818 _{±0.035}	0.821 _{±0.029}	0.825 _{±0.024}		RMSE	6.784 _{±0.339}	
	AUC	0.850 _{±0.034}	0.831 _{±0.017}	0.843 _{±0.029}	0.841 _{±0.019}	.001	MAE	4.815 _{±0.148}	.001
QCNN	ACC	0.856 _{±0.041}	0.789 _{±0.027}	0.832 _{±0.027}	0.826 _{±0.025}		RMSE	5.823 _{±0.308}	
	AUC	0.862 _{±0.025}	0.850 _{±0.015}	0.859 _{±0.024}	0.857 _{±0.014}	.001	MAE	3.662 _{±0.135}	.001
KGT	ACC	0.814 _{±0.034}	0.806 _{±0.041}	0.870 _{±0.024}	0.830 _{±0.023}		RMSE	6.191 _{±0.230}	
	AUC	0.871 _{±0.021}	0.846 _{±0.020}	0.875 _{±0.019}	0.864 _{±0.016}	.001	MAE	3.979 _{±0.236}	.001
JMMLRC	ACC	0.805 _{±0.042}	0.799 _{±0.038}	0.830 _{±0.025}	0.811 _{±0.028}		RMSE	7.126 _{±0.345}	
	AUC	0.852 _{±0.027}	0.820 _{±0.024}	0.822 _{±0.038}	0.831 _{±0.022}	.001	MAE	5.038 _{±0.168}	.001
DenseMoE	ACC	0.843 _{±0.035}	0.822 _{±0.020}	0.839 _{±0.026}	0.835 _{±0.021}		RMSE	5.989 _{±0.224}	
	AUC	0.857 _{±0.021}	0.839 _{±0.026}	0.855 _{±0.026}	0.850 _{±0.017}	.001	MAE	3.662 _{±0.244}	.001
MM-DLS	ACC	0.831 _{±0.031}	0.820 _{±0.025}	0.843 _{±0.030}	0.831 _{±0.021}		RMSE	5.374 _{±0.301}	
	AUC	0.869 _{±0.016}	0.848 _{±0.018}	0.857 _{±0.013}	0.858 _{±0.012}	.001	MAE	3.147 _{±0.126}	.001
Multi-Cog	ACC	0.842 _{±0.026}	0.825 _{±0.029}	0.860 _{±0.017}	0.842 _{±0.019}		RMSE	4.949 _{±0.199}	
	AUC	0.880 _{±0.024}	0.853 _{±0.019}	0.871 _{±0.022}	0.868 _{±0.010}	.005	MAE	3.092 _{±0.225}	.001
Ours (image)	ACC	0.890 _{±0.021}	0.855 _{±0.034}	0.894 _{±0.020}	0.880 _{±0.018}		RMSE	2.046 _{±0.197}	
	AUC	0.901 _{±0.029}	0.883 _{±0.017}	0.892 _{±0.026}	0.892 _{±0.019}	.005	MAE	1.618 _{±0.118}	.001
Ours (full)	ACC	0.899 _{±0.021}	0.872 _{±0.035}	0.905 _{±0.018}	0.892 _{±0.022}		RMSE	1.385 _{±0.082}	
	AUC	0.917 _{±0.025}	0.898 _{±0.018}	0.921 _{±0.027}	0.912 _{±0.018}	−	MAE	0.965 _{±0.142}	−

Table 3. Quantitative comparison of short-term treatment prognosis and long-term postoperative prognosis on the internal and external cohorts. Reported p -values are calculated against *Ours* (full) configuration.

Methods	Short-term treatment prognosis					Long-term postoperative prognosis		
	Metrics	TBUT _{IM}	CFS _{IM}	SIT _{IM}	OSDI _{IM}	$p <$	Metrics	PDE vs. NPD
<i>Internal validation</i>								
C-former	RMSE	7.383 _{±0.266}	7.123 _{±0.333}	7.534 _{±0.357}	7.282 _{±0.404}		ACC	0.820 _{±0.042}
	MAE	5.269 _{±0.274}	5.114 _{±0.145}	5.408 _{±0.302}	5.029 _{±0.186}	.001	AUC	0.893 _{±0.016}
MM-DLS	RMSE	6.876 _{±0.241}	6.565 _{±0.314}	7.168 _{±0.312}	6.726 _{±0.389}		ACC	0.835 _{±0.038}
	MAE	4.691 _{±0.248}	4.248 _{±0.136}	4.936 _{±0.283}	4.366 _{±0.165}	.001	AUC	0.916 _{±0.019}
JMMLRC	RMSE	6.315 _{±0.305}	6.107 _{±0.196}	6.814 _{±0.392}	7.263 _{±0.268}		ACC	0.845 _{±0.023}
	MAE	3.639 _{±0.124}	3.386 _{±0.213}	4.183 _{±0.167}	4.419 _{±0.244}	.001	AUC	0.908 _{±0.023}
DenseMoE	RMSE	6.364 _{±0.232}	6.044 _{±0.309}	6.649 _{±0.295}	6.201 _{±0.370}		ACC	0.845 _{±0.036}
	MAE	4.085 _{±0.237}	3.631 _{±0.126}	4.319 _{±0.273}	3.758 _{±0.156}	.001	AUC	0.923 _{±0.007}
QCNN	RMSE	7.107 _{±0.256}	6.890 _{±0.328}	7.395 _{±0.333}	6.954 _{±0.401}		ACC	0.823 _{±0.039}
	MAE	4.934 _{±0.261}	4.571 _{±0.139}	5.177 _{±0.295}	4.697 _{±0.168}	.001	AUC	0.901 _{±0.012}
Multi-Cog	RMSE	6.143 _{±0.319}	5.932 _{±0.204}	6.436 _{±0.406}	6.093 _{±0.277}		ACC	0.842 _{±0.024}
	MAE	3.764 _{±0.131}	3.310 _{±0.220}	4.011 _{±0.164}	3.538 _{±0.248}	.001	AUC	0.912 _{±0.021}
KGT	RMSE	6.217 _{±0.297}	5.505 _{±0.183}	6.903 _{±0.361}	7.364 _{±0.246}		ACC	0.850 _{±0.029}
	MAE	3.167 _{±0.115}	3.400 _{±0.208}	3.805 _{±0.142}	3.024 _{±0.223}	.001	AUC	0.923 _{±0.013}
MM-MIL	RMSE	6.124 _{±0.308}	5.808 _{±0.185}	7.132 _{±0.269}	6.867 _{±0.354}		ACC	0.857 _{±0.017}
	MAE	2.847 _{±0.121}	3.239 _{±0.206}	4.085 _{±0.243}	2.724 _{±0.135}	.001	AUC	0.929 _{±0.014}
Ours (image)	RMSE	3.247 _{±0.229}	3.186 _{±0.112}	4.392 _{±0.278}	4.639 _{±0.164}		ACC	0.892 _{±0.040}
	MAE	1.718 _{±0.084}	1.415 _{±0.168}	2.287 _{±0.102}	1.712 _{±0.174}	.001	AUC	0.958 _{±0.028}
Ours (full)	RMSE	2.472 _{±0.112}	2.352 _{±0.196}	3.643 _{±0.163}	3.714 _{±0.248}		ACC	0.914 _{±0.030}
	MAE	1.199 _{±0.167}	1.043 _{±0.045}	1.827 _{±0.184}	1.166 _{±0.056}	–	AUC	0.973 _{±0.015}
<i>External validation</i>								
C-former	RMSE	7.476 _{±0.395}	7.260 _{±0.262}	7.617 _{±0.466}	7.387 _{±0.324}		ACC	0.810 _{±0.028}
	MAE	5.416 _{±0.188}	5.175 _{±0.269}	5.360 _{±0.203}	5.091 _{±0.297}	.001	AUC	0.826 _{±0.042}
DenseMoE	RMSE	6.913 _{±0.359}	6.605 _{±0.226}	7.156 _{±0.443}	6.736 _{±0.294}		ACC	0.841 _{±0.035}
	MAE	4.753 _{±0.164}	4.313 _{±0.241}	4.814 _{±0.178}	4.434 _{±0.266}	.001	AUC	0.853 _{±0.018}
MM-MIL	RMSE	6.165 _{±0.220}	5.857 _{±0.298}	6.705 _{±0.297}	7.183 _{±0.373}		ACC	0.852 _{±0.019}
	MAE	3.680 _{±0.235}	3.436 _{±0.126}	4.139 _{±0.254}	4.369 _{±0.147}	.001	AUC	0.857 _{±0.030}
KGT	RMSE	6.408 _{±0.337}	6.092 _{±0.212}	6.646 _{±0.419}	6.219 _{±0.285}		ACC	0.838 _{±0.030}
	MAE	4.126 _{±0.155}	3.685 _{±0.234}	4.391 _{±0.170}	3.803 _{±0.258}	.001	AUC	0.864 _{±0.016}
MM-DLS	RMSE	7.157 _{±0.263}	6.846 _{±0.335}	7.390 _{±0.351}	6.978 _{±0.406}		ACC	0.822 _{±0.026}
	MAE	4.989 _{±0.276}	4.648 _{±0.159}	5.035 _{±0.294}	4.752 _{±0.172}	.001	AUC	0.835 _{±0.029}
JMMLRC	RMSE	6.194 _{±0.230}	5.882 _{±0.307}	6.429 _{±0.311}	6.015 _{±0.376}		ACC	0.843 _{±0.019}
	MAE	3.817 _{±0.238}	3.374 _{±0.127}	4.066 _{±0.271}	3.587 _{±0.143}	.001	AUC	0.869 _{±0.012}
Multi-Cog	RMSE	6.069 _{±0.314}	5.549 _{±0.181}	7.004 _{±0.378}	7.477 _{±0.256}		ACC	0.851 _{±0.017}
	MAE	3.494 _{±0.132}	3.651 _{±0.214}	3.541 _{±0.147}	3.070 _{±0.229}	.001	AUC	0.866 _{±0.020}
QCNN	RMSE	6.783 _{±0.216}	6.047 _{±0.302}	6.826 _{±0.271}	8.239 _{±0.403}		ACC	0.860 _{±0.024}
	MAE	3.349 _{±0.233}	3.971 _{±0.119}	3.283 _{±0.205}	2.802 _{±0.117}	.001	AUC	0.877 _{±0.013}
Ours (image)	RMSE	3.315 _{±0.231}	3.436 _{±0.124}	4.127 _{±0.263}	4.839 _{±0.184}		ACC	0.892 _{±0.033}
	MAE	2.584 _{±0.096}	2.315 _{±0.188}	1.762 _{±0.069}	1.815 _{±0.165}	.001	AUC	0.898 _{±0.019}
Ours (full)	RMSE	2.583 _{±0.115}	2.678 _{±0.206}	3.418 _{±0.146}	3.979 _{±0.265}		ACC	0.920 _{±0.021}
	MAE	1.805 _{±0.177}	1.770 _{±0.069}	1.076 _{±0.143}	1.176 _{±0.047}	–	AUC	0.916 _{±0.017}

Within each of the three proposed modules, components are introduced sequentially on top of the preceding configuration to examine their incremental contributions. In parallel, Individual Evaluation independently assesses each complete innovation under a common baseline rather than progressively stacking modules. Specifically, I0 uses a vanilla ResNet-50 visual backbone with a separate MLP classification/regression head for each task; I1 adds only relation-guided task-specific feature extraction to I0; I2 adds only patient-semantic-guided feature contextualization, with visual queries directly derived from the vanilla backbone; and I3 adds only patient-task-conditioned hypernetwork prediction. Thus, I1–I3 isolate the effect of each innovation without assistance from the other two modules. Overall, performance generally improves as the components are progressively introduced across diagnosis, regression, and prognosis.

Starting from the visual backbone (R0), R1 introduces clinically induced relation alignment, R2 further adds HSIC-based marginal screening under a fixed top- ρ budget, and R3 incorporates KCI-based conditional screening to complete task-specific robust feature filtering. A pronounced improvement is observed

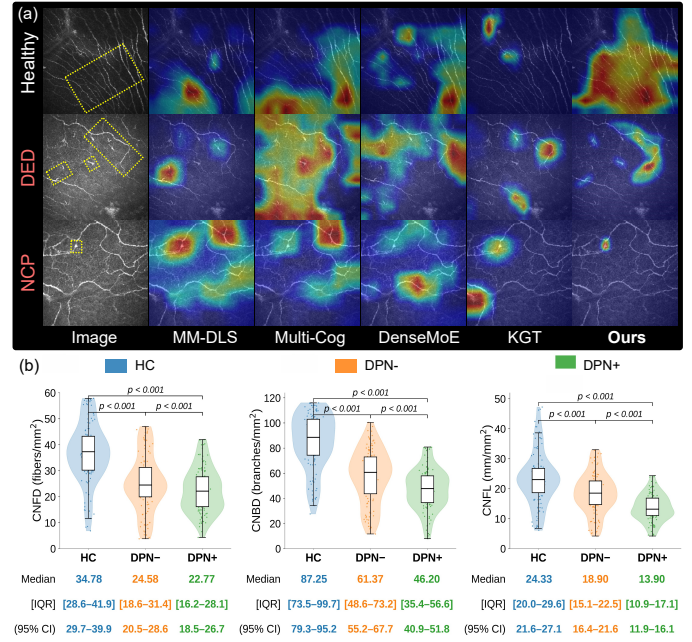


Fig. 4. Qualitative comparison and neuro-morphometric analysis. (a) Grad-CAM visualizations across representative CCM phenotypes. (b) Distributions of CNFD, CNBD, and CNFL across HC, DPN-, and DPN+.

for short-term treatment prognosis, where MAE decreases from 2.560 to 1.943, while long-term postoperative prognosis also improves from an AUC of 0.874 to 0.904. These consistent gains support the intended design in which clinically induced relation alignment regularizes nuisance-sensitive feature dependencies, while task-specific robust feature filtering retains evidence relevant to each prediction objective.

S0–S3 progressively introduce task-aware patient semantic encoding, patient-semantic-guided visual contextualization, and the residual connection on top of the task-specific visual representation. Systemic regression shows a substantial MAE reduction from 1.502 to 1.188, while systemic diagnosis exhibits a more modest but consistent AUC increase from 0.894 to 0.913. These improvements indicate that patient semantics provide complementary context for interpreting visually similar CCM patterns and reduce ambiguity that cannot be resolved from visual evidence alone.

H0–H3 progressively replace a static prediction head with task embedding, patient-task conditioning through the shared hypernetwork, and the task-specific parameter generator. The strongest representative improvement is observed for systemic regression, where MAE decreases from 1.188 to 0.820, whereas ocular diagnosis AUC increases from 0.927 to 0.950. The consistent improvements across different task types support patient-task-conditioned hypernetwork prediction in adapting the prediction function to both patient state and task identity rather than relying on a fixed mapping.

The individual evaluation further shows that each innovation provides a measurable standalone benefit. In particular, Semantic Contextualization alone yields only moderate improvements over I0, supporting that the patient-semantic branch does not obtain its gains from direct target leakage. More importantly, the modules exhibit complementary effects when combined: for

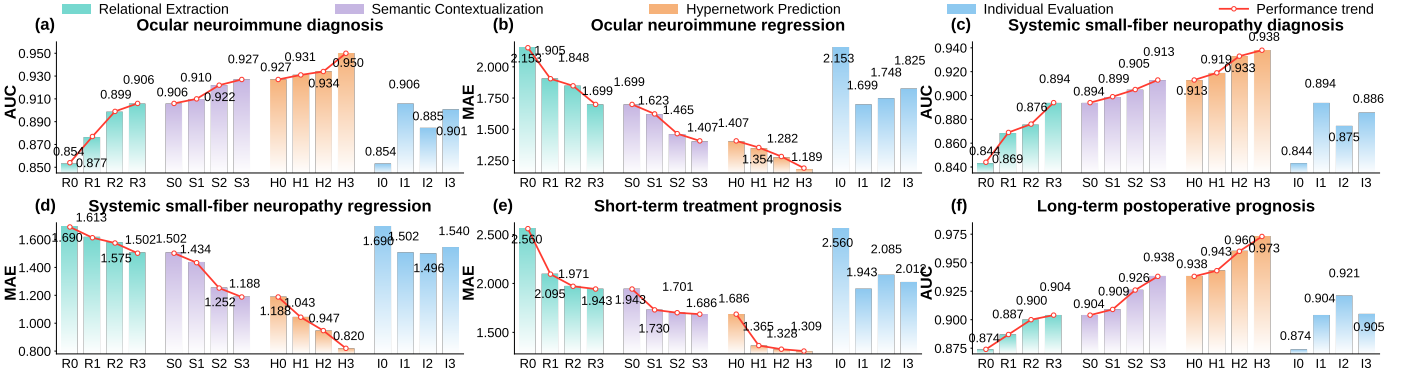


Fig. 5. Component-wise progressive ablation and individual evaluation on the internal validation cohort.

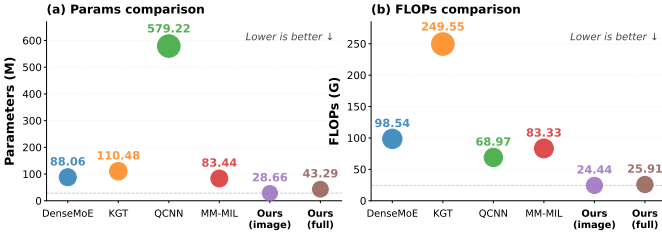


Fig. 6. Computational complexity comparison under *Ours (image)* and *Ours (full)* configurations.

systemic regression, Semantic Contextualization alone reduces MAE from 1.690 to 1.496, whereas introducing semantic contextualization after relation-guided feature extraction further reduces MAE from 1.502 to 1.188. Together with the subsequent improvement to 0.820 after Hypernetwork Prediction, these results indicate that the three innovations are mutually complementary and that their joint integration is critical to the overall performance gain.

4.4. Computational complexity analysis

Fig. 6 compares the computational complexity of the proposed framework under its full and image-only configurations. Under the complete-context setting, the *Ours (full)* configuration contains 43.29M parameters and requires 25.91G FLOPs, substantially fewer parameters than MM-MIL (83.44M) and KGT (110.48M). In the *Ours (image)* configuration, the ClinicalBERT-based clinical-context branch is removed, leaving relation-guided visual feature extraction and task-conditioned hypernetwork prediction with 28.66M parameters and 24.44G FLOPs. These results indicate that the performance gains are not driven by model-scale expansion; instead, the framework relies on shared representations and lightweight patient-task-conditioned parameter generation, yielding a favorable performance-complexity trade-off.

4.5. Hyperparameter sensitivity analysis

Overall validation performance was highest at $\lambda_{\text{prior}} = 1.0$, $\rho = 30\%$, and $\lambda_{\text{hyper}} = 5 \times 10^{-4}$ (Fig. 7). Across all task groups, performance generally improved toward intermediate settings and declined when the corresponding constraint became excessive. Specifically, intermediate λ_{prior} maintains a balance between relational guidance and visual flexibility, $\rho = 30\%$ retains

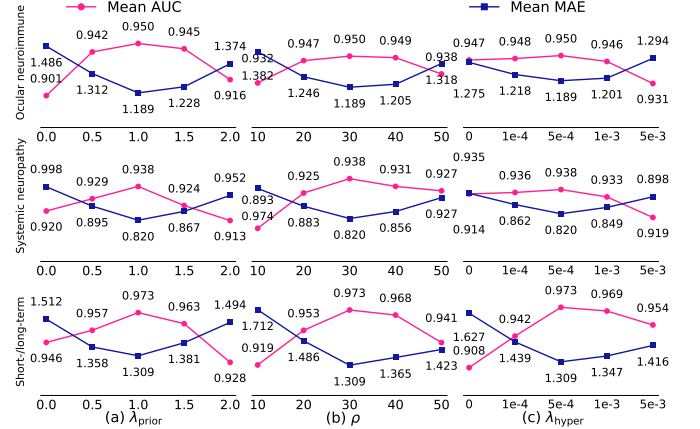


Fig. 7. Hyperparameter sensitivity across clinical tasks on the internal validation splits. Effects of λ_{prior} , ρ , and λ_{hyper} on ocular neuroimmune, systemic neuropathy, and short-/long-term performance.

informative task-specific factors while limiting redundant channels, and intermediate λ_{hyper} regularizes the generated parameters without overly restricting patient-task adaptation. These trends suggest a stable operating range for relation alignment, feature selection, and hypernetwork adaptation.

5. Discussion and Further Analysis

Beyond predictive performance, we further examined the mechanisms, representation structure, and clinical validity of the proposed framework. These analyses evaluate its behavior under task and acquisition variability, visualize the organization of disease-related representations, and assess whether the predicted clinical metrics and longitudinal outcomes remain consistent with established clinical patterns.

5.1. Mechanism and robustness analyses

Fig. 8 examines three properties of the proposed framework: whether clinically induced relations provide meaningful rather than generic regularization, whether patient-task adaptation reduces interference among heterogeneous objectives, and whether relation-guided feature extraction improves robustness to acquisition degradation.

To determine whether the gain from relation-guided feature extraction depends on clinically structured relations, Fig. 8(a)

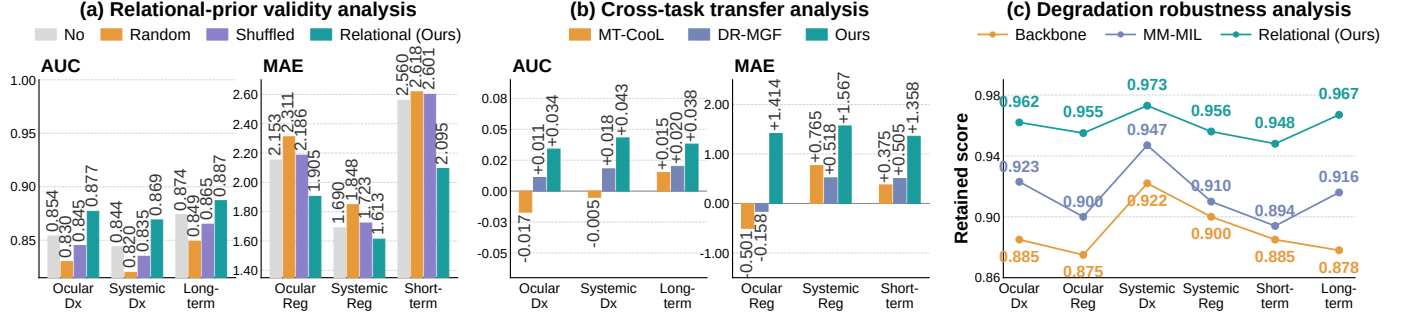


Fig. 8. Mechanism and robustness analyses. (a) Validity of the clinically induced relation structure across diagnosis, regression, and prognosis tasks. (b) Cross-task transfer relative to single-task learning. (c) Performance retention under input degradation for the backbone, MM-MIL, and relational feature extraction.

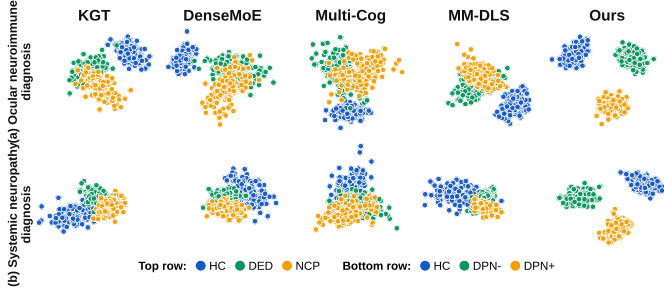


Fig. 9. t-SNE visualization of learned representations. Top: ocular neuroimmune diagnosis; bottom: systemic neuropathy diagnosis.

compares four settings. The *No* setting removes relation alignment, whereas *Random* replaces the clinical relation structure with an independently sampled symmetric matrix and *Shuffled* disrupts variable correspondence while preserving its value distribution; *Relational (Ours)* uses the clinically induced relations defined in Sec. 2.1.2. All settings were repeated using 10 independent random seeds, and the reported values represent the mean across runs. The clearest effect is observed for short-term treatment prognosis, where MAE decreases from 2.560 without relation guidance to 2.095 with the clinically induced structure, while both random and shuffled relations fail to reproduce this improvement. Similar trends are observed across the remaining objectives. This contrast indicates that the benefit is not explained by imposing an arbitrary relational constraint; rather, preserving clinically supported dependencies provides a more informative organization of the latent visual space.

Fig. 8(b) evaluates whether the proposed multi-task design alleviates interference among diagnosis, regression, and prognosis objectives. For each evaluated method, *Single* denotes six task-specific counterparts obtained by keeping the same architecture but retaining only one prediction head per model, so that each model performs a single task without cross-task interaction. Cross-task transfer is then computed against the corresponding single-task counterpart as $\Delta = \text{Method} - \text{Single}$ for AUC-based tasks and $\Delta = \text{Single} - \text{Method}$ for MAE-based tasks, where positive Δ indicates beneficial transfer. MT-Cool (Wu et al., 2025) and DR-MGF (Sun et al., 2026) are specifically designed to mitigate task interference and gradient conflict in multi-task learning, yet negative transfer remains on three and one of the six tasks, respectively. In contrast,

our framework achieves positive transfer across all six tasks, with gains of +0.034–+0.043 for AUC-based tasks and +1.358–+1.567 for MAE-based tasks. This consistent transfer supports the proposed design, in which task-specific feature selection limits inappropriate representation sharing and patient–task conditioning further adapts the prediction function to heterogeneous clinical objectives.

Representation stability under simulated acquisition degradation is further examined in Fig. 8(c), including blur, noise, illumination variation, and contrast perturbation. Performance retention is defined as $R = (M_{\text{deg}} - 0.5) / (M_{\text{clean}} - 0.5)$ for AUC-based tasks and $R = M_{\text{clean}} / M_{\text{deg}}$ for MAE-based tasks, such that higher values indicate better preservation relative to the corresponding performance baseline. A representative example is ocular regression, for which the retained score increases from 0.875 with the backbone and 0.900 with MM-MIL to 0.955 with *Relational (Ours)*. The same ordering is observed across the six prediction tasks. These results are consistent with the intended role of clinically induced relation alignment: constraining feature dependencies toward stable clinical structure reduces reliance on degradation-sensitive visual correlations and improves representation stability under suboptimal imaging conditions.

5.2. t-SNE visualization

Fig. 9 visualizes the organization of disease-related representations beyond the final prediction scores. Across both ocular neuroimmune and systemic neuropathy diagnosis tasks, the proposed framework produces more compact intra-class clusters and clearer inter-class separation than KGT, DenseMoE, Multi-Cog, and MM-DLS. The difference is particularly evident for clinically adjacent phenotypes: DED and NCP remain substantially overlapped for several competing methods, whereas the proposed framework shows clearer separation among HC, DED, and NCP. A similar reduction in overlap is observed among HC, DPN-, and DPN+. These visualization patterns are consistent with the intended roles of task-specific feature extraction and patient-semantic contextualization and provide qualitative evidence of more discriminative phenotype representations.

5.3. Clinical validity of metric regression and prognosis

Beyond numerical accuracy, we examined whether the predictions reproduced clinically expected patterns (Fig. 10). All

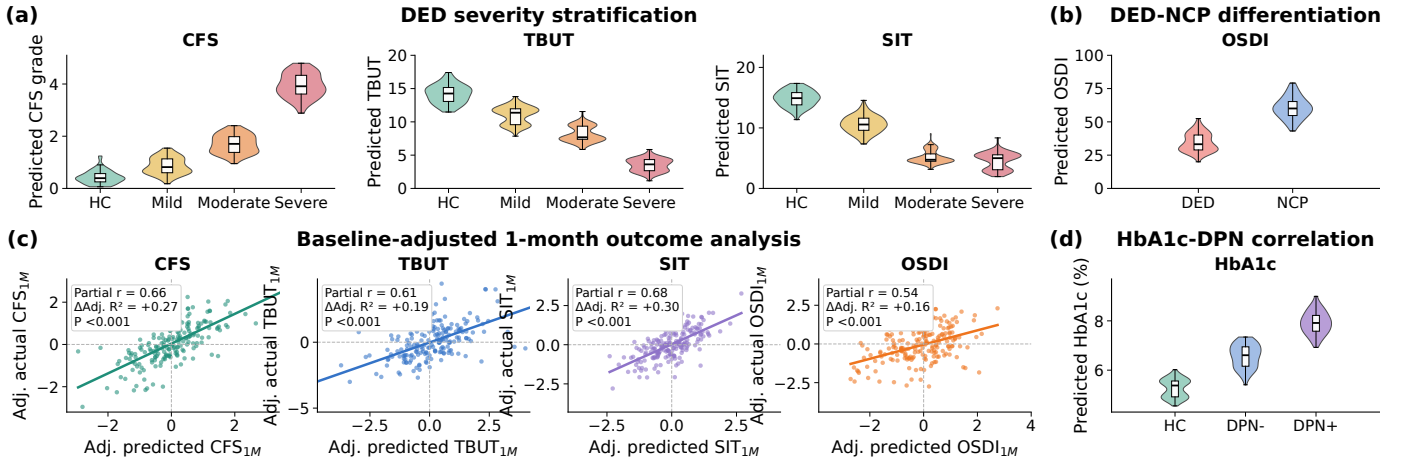


Fig. 10. Clinical validity of metric regression and prognosis. Associations with disease severity, phenotype differentiation, longitudinal outcomes, and systemic metabolic status are evaluated.

analyses used held-out predictions pooled from the five internal test folds and the complete corresponding external cohorts. To reduce potential confounding between patient context and clinical grouping variables, analyses involving disease severity or diagnostic status used the *Ours (image)* configuration, whereas the longitudinal analysis used *Ours (full)*.

For DED severity, we focused on CFS, TBUT, and SIT as complementary objective ocular-surface signs reflecting corneal epithelial staining, tear-film stability, and aqueous tear production, respectively. Predicted CFS increased progressively from HC to severe DED, whereas TBUT and SIT decreased, consistent with their established associations with DED severity (Sullivan et al., 2010; Begley et al., 2003) (Fig. 10(a)). Focusing on these three objective signs also avoids mixing patient-reported symptom burden with the assessment of ocular-surface changes.

OSDI was therefore examined separately as a patient-reported measure of ocular symptoms and vision-related functional impact. Predicted OSDI was substantially higher in NCP than in conventional DED, consistent with previous evidence of greater OSDI-assessed symptom burden in NCP (Chin et al., 2024) (Fig. 10(b)). Together, panels (a) and (b) show that the image-derived predictions preserve clinically meaningful patterns across both objective ocular-surface signs and patient-reported symptoms.

We further evaluated whether the predicted one-month outcomes retained information beyond the corresponding baseline measurements (Fig. 10(c)). All four endpoints—CFS, TBUT, SIT, and OSDI—were included here to assess complementary objective signs and patient-reported symptoms over follow-up. For each endpoint, the observed one-month value y_{1M} was modeled using the baseline value y_0 and the held-out model prediction $\hat{y}_{1M}$, i.e., $y_{1M} = \beta_0 + \beta_1 y_0 + \beta_2 \hat{y}_{1M} + \epsilon$. Partial correlations were computed after adjustment for y_0 , while the increase in adjusted R^2 over the baseline-only model quantified the additional explanatory value of the prediction. All four endpoints remained significantly associated with their observed one-month outcomes after baseline adjustment, indicating that the full model captured longitudinal information beyond base-

line status.

Predicted HbA1c increased from HC to DPN- and DPN+, consistent with the established association between greater glycemic exposure and diabetic peripheral neuropathy (Braf-fett et al., 2020; Pinto et al., 2020) (Fig. 10(d)). Collectively, these analyses show that the predicted clinical metrics retain clinically meaningful cross-sectional and longitudinal patterns. These findings further support the clinical relevance of integrating patient context into individualized outcome prediction.

6. Conclusion

We addressed the challenge of unified CCM-based assessment across ocular neuroimmune and systemic neuropathy diagnosis, continuous clinical metric regression, and longitudinal prognosis. The proposed framework combines relation-guided task-specific feature extraction, patient-semantic-guided feature contextualization, and patient-task-conditioned hypernetwork prediction to learn robust task-relevant representations while adapting prediction to individual patients and heterogeneous clinical objectives. Across internal and independent external patient cohorts, the framework achieved consistently strong performance across categorical, continuous, and prognostic tasks, while the predicted clinical metrics preserved clinically meaningful disease-related patterns. These findings support the potential of CCM-derived visual biomarkers for disease assessment, quantitative clinical characterization, and longitudinal prediction, with patient context providing optional complementary information.

A key limitation is that the clinical objectives were evaluated in task-specific cohorts rather than in a single cohort undergoing the complete set of assessments. Future work should therefore focus on prospective validation in unified, more heterogeneous patient cohorts to further assess performance across the complete clinical assessment pathway.

7. Acknowledgement

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